MITOCHONDRIAL & CELLULAR ENERGY

Renew — Full Review

FORMULA

MOTS-c 10mg + SS-31 10mg + NAD+ 500mg

What's in it

PEPTIDE	DOSE	ROLE IN THIS BLEND
MOTS-c	10mg	The signal piece. Activates AMPK, the cell's main metabolic switch, telling mitochondria to multiply and burn fuel more efficiently.
SS-31	10mg	The structural piece. Binds and stabilizes cardiolipin in the inner mitochondrial membrane — the only ingredient here with a real FDA approval behind it (as elamipretide, for Barth syndrome).
NAD+	500mg	The fuel piece. The coenzyme every one of those energy-producing reactions actually runs on — levels decline substantially with age.

The combination logic

Each of these three works on a different part of the same problem. SS-31 repairs the physical structure inside the mitochondria where energy is made. MOTS-c sends a signal that tells the cell to build more mitochondria and use fuel more efficiently. NAD+ supplies the raw material every one of those energy-producing reactions actually runs on. Put together, the idea is structure, signal, and fuel, addressed at the same time. Clinical detail: SS-31 (elamipretide) selectively binds cardiolipin in the inner mitochondrial membrane, protecting cristae architecture and electron transport chain function. MOTS-c is a mitochondrial-DNA-encoded peptide that activates AMPK signaling, promoting mitochondrial biogenesis and improved glucose/fatty-acid metabolism. NAD+ is the essential

coenzyme substrate for Complex I of the electron transport chain and for sirtuin activity; its cellular pool declines significantly with age. The three are proposed to act on non-competing points of the same energy-production pathway.

Is there evidence for the combination, or only for the parts? SS-31 has the most developed evidence of the three, including FDA accelerated approval (as elamipretide) for a rare mitochondrial condition, Barth syndrome — though that approval does not extend to general energy or longevity use. MOTS-c has meaningful laboratory research on metabolic adaptation and preliminary human interest. NAD+ has a large body of general metabolic literature but is a coenzyme, not a drug, and dosing conventions vary widely by route. No study has tested this specific three-part combination.

What it's used for

USED FOR	IN SHORT
Cellular energy / fatigue support	The most consistent rationale across the three components — each targets a distinct point in mitochondrial energy production.
Metabolic and age-related decline Mitochondrial membrane protection	MOTS-c in particular has been studied for metabolic adaptation and insulin sensitivity in laboratory research. SS-31 (elamipretide) has the most developed human data of the three, including an FDA-approved use for a rare mitochondrial disease.

Overlaps & stacked risks

- Each component carries its own risk profile; combining them does not reduce that.
- MOTS-c and NAD+ both interact with glucose/insulin metabolism — monitor in patients with diabetes.
- Rapid NAD+ administration is the most common source of acute discomfort (flushing, chest tightness) in this combination.
- Long-term human safety data for this specific combination does not exist.

Who it is not for

- Pregnant or breastfeeding patients
- Patients with significant cardiac or renal disease, without clinician oversight
- Anyone on blood-sugar-lowering medication, due to MOTS-c’s effect on glucose metabolism

Patients expecting a fast-infusion NAD+ experience — rapid administration can cause transient flushing or chest tightness

Who it suits

Patients pursuing general energy, metabolic, or age-related support who have discussed the individual evidence profile of each component (strongest for SS-31, more preliminary for MOTS-c and NAD+ dosing) with their provider.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller. That designation carries specific, enforceable obligations:

A prescription in your name. Compounded from a prescription for one identified patient, after an evaluation.

Licensed and inspected. A state licence, subject to inspection by the state board of pharmacy.

Ingredients that can be traced. A certificate of analysis from an FDA-registered facility, back to a named source and lot.

Sterile preparation to a published standard. USP standards.

A paper trail. A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website. This page describes Renew for general educational purposes. It is not medical advice and does not replace a consultation with a licensed provider. Individual results vary, and combination products carry the risks of each included peptide.

IMPORTANT SAFETY INFORMATION

Compounded medications offered through FitClinicMD are prescribed only after an individual clinical evaluation by a licensed provider, and are prepared by state-licensed compounding pharmacies. Compounded medications have not been evaluated or approved by the U.S. Food and Drug Administration as safe or effective for the treatment of any condition. Information on this site is for general educational purposes and is not medical advice. You must be 18 or older. Individual results vary. A prescription is issued only if a licensed provider determines it is appropriate for you.